Fast Statistical Detection, Segmentation and Classification of Moving Objects by Sufficient-Statistic Window Scanning

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November 4, 2026
Preprint, first full draft

Abstract

Moving objects in video are usually found with cheap background subtraction, which returns pixels but no classes and no error guarantee, or with trained deep detectors, which need labelled data and a good deal of computation per frame. We study a purely statistical alternative, SWIFT-MO. A per-pixel adaptive Gaussian background gives a motion statistic; integral images of its sufficient statistics make the score of *every* sub-window of every size computable in constant time, so all windows on a grid are scanned in each frame. The alarm threshold is an order statistic of the maximum of the scan statistic over object-free frames, which bounds the false-alarm probability per frame, jointly for all windows, in finite samples and without distributional assumptions, provided object-free frames are exchangeable. Detected windows are tracked and classified against Gaussian colour models by Wald's sequential probability ratio test. We prove that the window likelihood computed from integral images is exact and that the threshold has the stated guarantee, and we evaluate the method on synthetic scenes with exact ground truth, on ten scenes that were not used during development. SWIFT-MO scores 84,550 windows per 320×240 frame in 5.3 ms on two CPU cores and raised no false alarm in 1000 object-free frames against a target of 0.05, a result we show to come from a conservative threshold, because the guarantee itself holds only after the background has settled (Section 5.6); its detection F_1 is 0.82, slightly below the 0.85 of OpenCV's MOG2 and KNN subtractors, which run about three times faster and give no class labels. A calibration study on 30 further streams finds the achieved false-alarm probability to be 0.000, 0.100 and 0.045 when the threshold is calibrated after 0, 100 and 300 burn-in frames (target 0.05), as the finite-sample theory of Section 4 predicts once the frames are close to exchangeable and not before. The calibrated threshold and the illumination compensation are necessary on these data (a fixed threshold alarms in every frame); the sequential classifier did not improve accuracy. No real video and no trained detector have been used yet, so the paper reports a method with proofs and a first, limited, synthetic evaluation.

Keywords: moving-object detection; scan statistics; integral images; sequential probability ratio test; distribution-free inference; real-time video analysis.

1 Introduction

Video analysis on cameras, drones and other small devices needs three answers per frame: where the moving objects are, which pixels belong to them, and what they are. Two routes are common, and each gives something up. Background subtraction is cheap, but it returns moving pixels only, it carries no class labels, and its threshold is tuned by hand, so nothing is known

about how often it raises a false alarm. Deep detectors return boxes and classes, but they must be trained on labelled data for each application, and on a CPU their cost is large against the time budget of one frame.

We ask whether the three tasks can be served by one statistical engine that needs no network training, runs on a CPU, and comes with a stated error rate. The starting point is simple. If pixels are modelled by Gaussian distributions, the log-likelihood of a window depends on the pixels only through their number, their sum and their sum of products. Those are sums over a rectangle, and rectangle sums are available in constant time from integral images [1, 9]. Hence every sub-window of every size and aspect ratio can be scored, and the search is limited by the number of windows, not by the cost of scoring one.

Once all windows are scored, detection is a multiple-testing problem over many thousands of strongly dependent windows. We treat it as such: the threshold is calibrated on the maximum of the scan statistic over object-free frames, so the false-alarm probability per frame is bounded for all windows at once. Classification is sequential: the evidence for each class is accumulated along the frames of a track and a decision is taken when it crosses Wald's boundary [11].

Contributions.

(1) An exact constant-time computation of the Gaussian log-likelihood of any window from integral images of sufficient statistics (Proposition 1).

(2) A scan statistic for moving objects built on an adaptive background, with a finite-sample, distribution-free bound on the false-alarm probability per frame over all windows jointly (Proposition 2).

(3) A statistical treatment of the scan: the motion statistic as an exact likelihood-ratio statistic, its variance under spatial dependence, a localisation property, and the exact distribution of the achieved false-alarm probability (Section 4), with an empirical check of the last (Section 5.6).

(4) A tracking and sequential-classification stage whose decision delay is measured and whose error levels are set by the user.

(5) An open-source implementation with a simulator that gives exact ground truth, and a held-out evaluation that includes ablations, background-subtraction baselines and negative results.

What the paper does *not* show: any result on real video, any comparison with a trained deep detector, and any speed-up from temporal gating (a design that is described but not implemented). Section 6 states what would have to be done to support such claims.

2 Related work

Integral images and sliding windows. Summed-area tables [1] and their use for constant-time rectangle features in real-time detection [9] underlie our scoring step. We use them differently: not to compute hand-designed features that a classifier weighs, but to obtain sufficient statistics from which a likelihood is computed exactly.

Efficient search over windows. Efficient subwindow search [5] finds the best window by branch and bound without scoring all of them. We score all windows on a stride grid, because several objects may be present and every locally maximal window is needed.

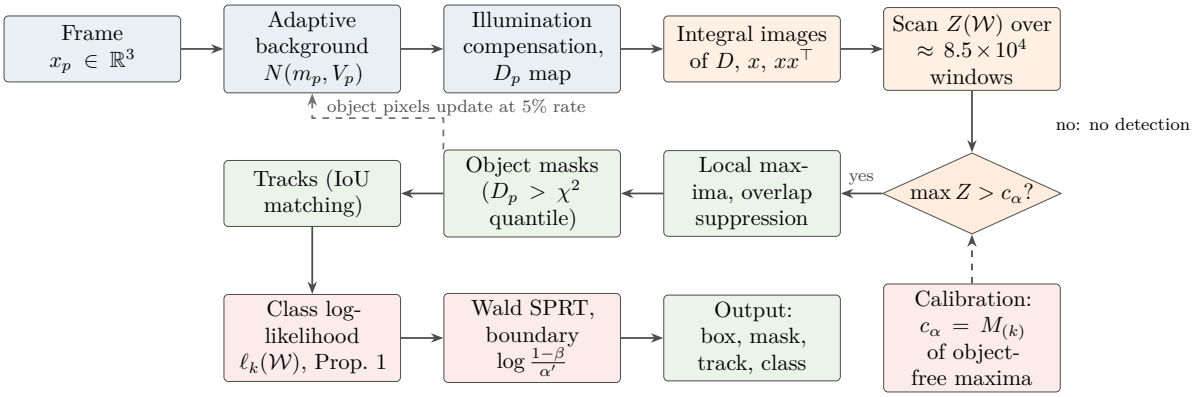


Figure 1: Pipeline of SWIFT-MO. Blue: per-pixel evidence; orange: scan of all windows, constant time each; green: segmentation and tracking; red: statistical decisions (class evidence and sequential test, and the calibrated threshold). Dashed: the threshold is set from object-free frames; masks feed back to the background update.

Scan statistics. The scan statistic, the maximum of a local statistic over windows, and its null distribution are classical [7, 4, 2], mostly for point processes in space and time. We apply the same logic to the pixel-wise motion statistic of a video frame and calibrate the maximum empirically.

Background subtraction. Adaptive Gaussian models per pixel and their mixtures [8] decide pixel by pixel. We pass the same per-pixel evidence to window statistics, so a decision is made on a region.

Distribution-free error control. Rank-based thresholds with finite-sample validity under exchangeability are the core of conformal inference [10, 6]. We use the simplest such threshold, an order statistic of null maxima.

Sequential testing. Wald's sequential probability ratio test [11] minimises the expected number of independent observations among tests with the same error levels; we apply it to the frames of a track, where independence holds only approximately.

Benchmarks. Change detection is commonly evaluated on CDnet 2014 [3]. We have not yet run it. A systematic literature search for a prior work that combines the four elements (exact window likelihoods from integral images, a conformal scan threshold on video frames, tracking, and sequential classification) is still to be done and could reduce the novelty claim.

3 Method

A frame is an $H \times W$ array of pixel vectors $x_p \in \mathbb{R}^d$ ($d = 3$ for colour). A *window* $\mathcal{W}$ is an axis-aligned rectangle with n pixels. Figure 1 shows the whole pipeline; the subsections follow its boxes from left to right.

3.1 Constant-time window likelihood

Model the pixels of an object class k as independent $N(\mu_k, \Sigma_k)$. For a window put $S_1 = \sum_{p \in \mathcal{W}} x_p$ and $S_2 = \sum_{p \in \mathcal{W}} x_p x_p^\top$.

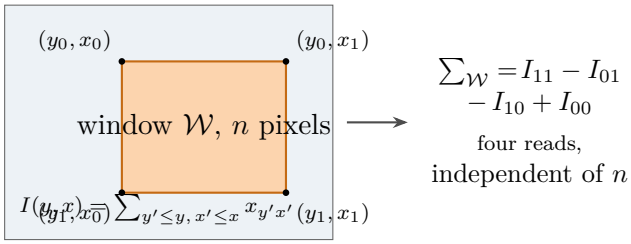


Figure 2: A rectangle sum from an integral image. The same four reads are used for $\sum x_p$, $\sum x_p x_p^\top$ and $\sum D_p$.

Proposition 1. *The log-likelihood of the pixels of $\mathcal{W}$ under class k depends on the data only through (n, S_1, S_2):*

$$\ell_k(\mathcal{W}) = -\tfrac{1}{2}\Big[n\big(d\log 2\pi + \log|\Sigma_k|\big) + \operatorname{tr}\big(\Sigma_k^{-1}(S_2 - S_1\mu_k^\top - \mu_k S_1^\top + n\mu_k\mu_k^\top)\big)\Big]. \tag{1}$$

If the integral images of the d channels x_i and of the $d(d+1)/2$ products $x_i x_j$ are built once per frame, then (n, S_1, S_2) of any window cost four table reads per channel, so $\ell_k(\mathcal{W})$ costs $O(d^2)$ operations regardless of n.

Proof. Sum the log-density over $\mathcal{W}$ and expand $(x-\mu)^\top\Sigma^{-1}(x-\mu) = x^\top\Sigma^{-1}x - 2\mu^\top\Sigma^{-1}x + \mu^\top\Sigma^{-1}\mu$. The sum of the quadratic form equals $\operatorname{tr}\big(\Sigma^{-1}\sum_p (x_p-\mu)(x_p-\mu)^\top\big)$, and $\sum_p (x_p-\mu)(x_p-\mu)^\top = S_2 - S_1\mu^\top - \mu S_1^\top + n\mu\mu^\top$. A rectangle sum from an integral image I is $I(y_1,x_1) - I(y_0,x_1) - I(y_1,x_0) + I(y_0,x_0)$. □

3.2 Motion scan statistic

Each pixel has an adaptive background $N(m_p, V_p)$ with diagonal V_p, updated by exponential forgetting with rate 0.02; pixels inside detected objects are updated at 5% of that rate. A global illumination change must not be mistaken for motion, so the per-channel median of the frame difference is subtracted before scoring and before the variance update. Let $D_p = (x_p - m_p)^\top V_p^{-1}(x_p - m_p)$, capped at a constant to limit the influence of outliers. On background D_p is approximately χ^2_d, and we define

$$Z(\mathcal{W}) = \frac{\sum_{p\in\mathcal{W}} D_p - nd}{\sqrt{2nd}}, \tag{2}$$

which has mean 0 and variance 1 for independent χ^2_d terms and no cap. One integral image of D gives $Z(\mathcal{W})$ in $O(1)$ for every window. Windows are taken on a grid of stride s, over a geometric set of sizes and three aspect ratios. For a compact object Z is largest at windows that match it, because added background pixels contribute zero mean and enlarge the denominator.

3.3 A calibrated threshold over all windows

The threshold has to hold for the maximum over the whole window set, not for one window. Let $M = \max_{\mathcal{W}} Z(\mathcal{W})$ on a frame without moving objects.

Proposition 2. *Let $M_1, \dots, M_m$ be the maxima of m object-free frames and M_{m+1} that of a new object-free frame. If $(M_1, \dots, M_{m+1})$ are exchangeable, and c is the k-th smallest of $M_1, \dots, M_m$ with $k = \lceil (m+1)(1-\alpha)\rceil \le m$, then*

$$\mathbb{P}(M_{m+1} > c) \le \alpha. \tag{3}$$

Proof. By exchangeability, and after breaking ties at random, the rank of M_{m+1} among the $m+1$ values is uniform on $\{1,\dots,m+1\}$. $M_{m+1} > c$ implies that its rank exceeds k, which has probability $(m+1-k)/(m+1) \le \alpha$. □

The bound needs $m \ge \lceil 1/\alpha \rceil - 1$ calibration frames and holds whatever the number of windows, their overlap and the distribution of the pixels. Its assumption is that object-free frames are exchangeable. With an adaptive background this holds only approximately: the first frames after initialisation are *not* exchangeable with later ones, and the calibration study of Section 5.6 shows the consequence, namely a conservative threshold when calibrated early and an anti-conservative one when calibrated after a short burn-in.

3.4 Segmentation, tracking and sequential classification

All locally maximal windows above the threshold (maximum of Z over a window-sized neighbourhood at each size) are collected. A greedy suppression keeps the best window and removes every window that overlaps a kept one by at least half of the smaller area, which removes both sub-windows of an object and windows that cover several objects. Inside a kept window, pixels with D_p above a χ^2 quantile form the object mask.

Detections are matched to existing tracks by intersection over union. For a track and a frame, the class log-likelihoods of the mask pixels, computed by (1), are scaled to $n_{\text{eff}} = 20$ effective independent pixels (neighbouring pixels are dependent) and added to the track's accumulated evidence $A(k)$. With $\lambda(k) = A(k) - \log \sum_{j \ne k} e^{A(j)}$ the log-odds of class k against the rest, class k is declared when $\lambda(k) \ge \log \frac{1-\beta}{\alpha'}$, Wald's upper boundary with error levels $\alpha' = \beta = 0.01$.

3.5 Complexity

Per frame the integral images cost $O\big(HW(1 + d + d(d+1)/2)\big)$, and each window then costs $O(1)$. With stride s, S sizes and A aspect ratios there are about $(HW/s^2)SA$ windows. For 320×240, $s = 4$ and 27 size–shape pairs we score 84,550 windows. Scanning all windows with $s = 1$ and all sizes would need about $(HW)^2/4$ windows, which is not feasible in real time.

4 Statistical foundations

This section fixes the objects of the method as statistical definitions and states what can be proved about them. Throughout, χ^2_ν is the chi-square law with ν degrees of freedom and $\mathbb{P}_0$ the law of an object-free frame.

Definition 1 (Window family and scan statistic)**.** Let $\mathcal{F}$ be a finite set of N windows (rectangles on a stride grid, over a set of sizes and aspect ratios). For a frame with per-pixel statistics D_p the *scan statistic* is $M = \max_{\mathcal{W} \in \mathcal{F}} Z(\mathcal{W})$, with Z as in (2).

Definition 2 (Null hypothesis and false-alarm probability)**.** H_0: the frame contains no moving object, and the pixels follow the background, $x_p \sim N(m_p, V_p)$. For a threshold c the *false-alarm probability per frame* is $\mathrm{FAP}(c) = \mathbb{P}_0(M > c)$, the probability that at least one of the N windows exceeds c; it is a family-wise error rate over $\mathcal{F}$.

Proposition 3 (Motion statistic as a likelihood-ratio statistic)**.** *Let V_p be known and let the alternative allow every pixel in $\mathcal{W}$ its own mean. Then the likelihood-ratio statistic of H_0 against it is*

$$-2 \log \Lambda(\mathcal{W}) = \sum_{p \in \mathcal{W}} D_p,$$

and under H_0 with independent pixels $-2 \log \Lambda \sim \chi^2_{nd}$ exactly, so $Z(\mathcal{W})$ has mean 0, variance 1 and skewness $\sqrt{8/(nd)}$.

Proof. The maximum-likelihood mean of pixel p under the alternative is x_p, which makes its log-density $-\frac{1}{2}\log|2\pi V_p|$; under H_0 it is lower by $\frac{1}{2}D_p$. Sum over $\mathcal{W}$. Each $D_p \sim \chi^2_d$ under H_0, independent across pixels, so the sum is χ^2_{nd}, with mean nd, variance $2nd$ and skewness $\sqrt{8/(nd)}$. □

Thus $\sum D_p$ is not an ad-hoc score: it is the generalised likelihood-ratio statistic, and the standardisation (2) is exact for the moments. The null law of Z is nearly normal only for large nd, which is one more reason to calibrate c on the *maximum* instead of using a normal quantile.

Proposition 4 (Sufficiency)**.** *For independent pixels $N(\mu, \Sigma)$ in a window, (n, S_1, S_2) is a sufficient statistic for (μ, Σ), and minimal sufficient for n fixed.*

Proof. By (1) the joint density factorises as $h(x)\, g_{\mu,\Sigma}(S_1, S_2)$ with $h \equiv (2\pi)^{-nd/2}$; the Fisher–Neyman criterion gives sufficiency. Two samples with the same likelihood ratio for all (μ, Σ) have the same (S_1, S_2) (the exponential family is full rank), which gives minimality. □

This is why the integral images lose no information about the Gaussian window likelihood, for classification as for detection.

Proposition 5 (Variance under spatial dependence)**.** *Let the D_p in $\mathcal{W}$ have the marginal law χ^2_d and average pairwise correlation $\bar\rho = \frac{1}{n(n-1)}\sum_{p\neq q}\mathrm{Corr}(D_p, D_q)$. Then*

$$\mathrm{Var}\, Z(\mathcal{W}) = 1 + (n-1)\bar\rho, \qquad n_{\mathrm{eff}} = \frac{n}{1+(n-1)\bar\rho} \xrightarrow{n\to\infty} \frac{1}{\bar\rho}.$$

Proof. $\mathrm{Var}\sum_p D_p = \sum_p \mathrm{Var}\, D_p + \sum_{p\neq q}\mathrm{Cov}(D_p, D_q) = 2nd + n(n-1)\bar\rho\, 2d$; divide by $2nd$. □

With $\bar\rho > 0$, the nominal normal quantiles of Z are too small, and increasingly so for large windows; this is the cause of the failure of the fixed threshold $Z > 3$ that we observe, and it is the reason why the effective sample size used in the sequential test is set to $n_{\mathrm{eff}} = 20$ instead of the number of mask pixels (an average correlation of about 0.05 corresponds to $n_{\mathrm{eff}} \to 20$).

Proposition 6 (Localisation)**.** *Suppose an object occupies a set O of $|O|$ pixels on which $\mathrm{E}\, D_p = d + \delta$ with $\delta > 0$, and $\mathrm{E}\, D_p = d$ elsewhere. Then for every window, $\mathrm{E}\, Z(\mathcal{W}) \le \delta\sqrt{|O|/(2d)}$, with equality if and only if $\mathcal{W} = O$.*

Proof. $\mathrm{E}\, Z(\mathcal{W}) = \delta|\mathcal{W}\cap O|/\sqrt{2d|\mathcal{W}|}$. Since $|\mathcal{W}\cap O| \le \min(|\mathcal{W}|, |O|)$, we have $|\mathcal{W}\cap O|/\sqrt{|\mathcal{W}|} \le \sqrt{|\mathcal{W}\cap O|} \le \sqrt{|O|}$, and both inequalities are equalities only if $\mathcal{W} \subseteq O$ and $O \subseteq \mathcal{W}$. □

Figure 3 shows the mean of Z for nested windows around one object: it rises up to the object, then falls as background pixels dilute it. This justifies selecting local maxima of Z across window sizes as object boxes, and shows that the statistic is largest for a window that equals the object, as far as the grid allows.

Proposition 7 (Exact law of the achieved false-alarm probability)**.** *Let $M_1, \ldots, M_m$ be i.i.d. with continuous distribution function F, let $c = M_{(k)}$ be the k-th order statistic and $U = 1 - F(c)$ the false-alarm probability that c actually achieves on a fresh frame. Then $U \sim \mathrm{Beta}(m+1-k,\, k)$, with mean $(m+1-k)/(m+1) \le \alpha$ for $k = \lceil (m+1)(1-\alpha)\rceil$.*

Proof. $F(M_{(k)})$ is the k-th order statistic of m uniform variables, which has the law $\mathrm{Beta}(k, m+1-k)$; take $1-$ of it. □

Proposition 2 bounds only the *average* of U over calibration samples. A single calibration sample gives a U that varies: for $m = 100$ and $\alpha = 0.05$ we have $k = 96$ and $U \sim \mathrm{Beta}(5, 96)$, with mean 0.0495, standard deviation 0.0215 and 5%/95% quantiles 0.020/0.089. Calibrating on more frames narrows this; the relative standard deviation is about $1/\sqrt{\alpha m}$. Section 5.6 compares these values with measurements.

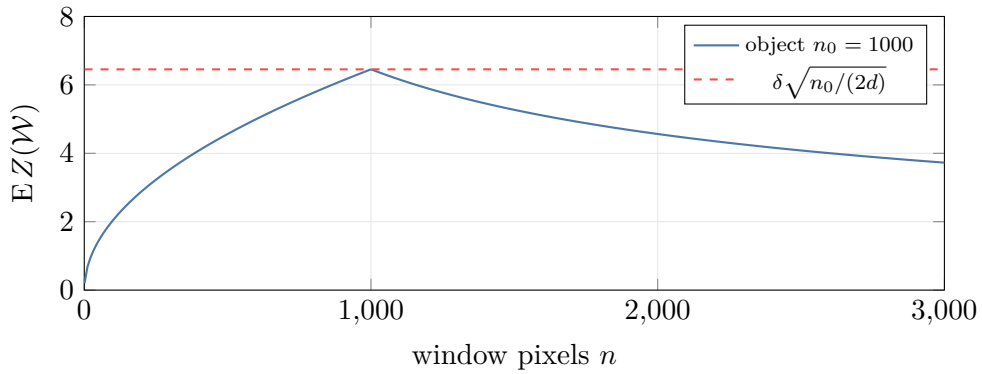


Figure 3: Proposition 6 for $d = 3$, an object of $n_0 = 1000$ pixels and mean excess $\delta = 0.5$ of D_p over its null mean. Inside the object $\mathrm{E}\, Z \propto \sqrt{n}$; once the window is larger than the object, $\mathrm{E}\, Z \propto n^{-1/2}$.

Separation	KL (closest pair)	KL (mean pair)	Drift per frame	Predicted E[N]	Observed
1	10.434	12.121	208.7	1.00	1.01
0.25	0.652	0.758	13.0	1.00	1.11
0.1	0.104	0.121	2.1	2.20	3.91

Table 1: Predicted expected decision time of the sequential test, $\max\{1, A/(n_{\text{eff}}\mathrm{KL})\}$ with the KL of the closest pair of classes, against the observed mean number of frames (Table 4). Per-pixel colour sd is 22, $n_{\text{eff}} = 20$, $A = \log 99$.

Proposition 8 (Sequential test: error and delay)**.** *Let the decision rule stop when the log-odds first crosses $A = \log \frac{1-\beta}{\alpha'}$ (declare) or $-B$, $B = \log \frac{1-\alpha'}{\beta}$ (reject). For i.i.d. increments Wald's inequalities give actual error levels $\alpha_{\mathrm{a}} \le \alpha'/(1-\beta)$, $\beta_{\mathrm{a}} \le \beta/(1-\alpha')$ and $\alpha_{\mathrm{a}} + \beta_{\mathrm{a}} \le \alpha' + \beta$, and the expected number of frames under the true class is approximately*

$$\mathrm{E}\, N \approx \frac{A}{n_{\text{eff}}\, \mathrm{KL}}, \qquad \mathrm{KL} = \tfrac{1}{2}(\mu_k - \mu_j)^{\top} \Sigma^{-1} (\mu_k - \mu_j)$$

per pixel, for Gaussian classes with common covariance (ignoring the overshoot over the boundary).

The Kullback–Leibler divergence between $N(\mu_k, \Sigma)$ and $N(\mu_j, \Sigma)$ is half the squared Mahalanobis distance of the means, and the accumulated log-likelihood ratio drifts by $n_{\text{eff}}\mathrm{KL}$ per frame. Table 1 compares this approximation with the decision times in Table 4. The approximation is in the right range for the two easy separations, where the first frame already crosses the boundary (one frame at least), and gives 2.2 against 3.9 at the hardest one; the difference is expected from the overshoot, from the dependence between frames, and from track and mask errors that perturb the increments, none of which the approximation includes. The error levels of the test are not guaranteed here, since the frames of a track are dependent and the scaling n_{eff} is a device.

Evaluation measures. A detection and a true box match if their intersection over union is at least 0.5, each true box being used once. With TP, FP, FN the numbers of matched detections, unmatched detections and unmatched true boxes: precision $= TP/(TP + FP)$, recall $= TP/(TP + FN)$, $F_1 = 2TP/(2TP + FP + FN)$. The empirical false-alarm rate is the share of object-free frames with at least one detection. Comparisons between methods use the *same* scenes, so we report paired differences with a t-interval and the Wilcoxon signed-rank test, which does not assume normal differences; with ten scenes the smallest possible two-sided p is

0.002. When no false alarm is observed in n trials, the one-sided 95% Clopper–Pearson upper bound on the probability is $1 - 0.05^{1/n} \approx 3/n$ (the "rule of three"): for $n = 1000$ it is 0.003, but the frames of one stream are dependent, so it is a bound for the stream, not for independent frames.

5 Experiments

All experiments are on synthetic scenes with exact ground truth. They check the method against its own assumptions; they are not evidence of performance on real video.

5.1 Setup

The simulator renders 320×240 colour frames: a smooth random background, additive Gaussian pixel noise (sd 5), a slow sinusoidal illumination drift (amplitude 6 grey levels), and three elliptical objects (28 to 64 pixels across) that bounce at constant speed. Objects belong to three classes (red, green, yellow) whose pixels are independent Gaussian with overlapping colour distributions (per-channel sd 22). A parameter, the class separation, pulls the three class means towards their common mean: 1 is the default, 0.25 and 0.1 make the classes much harder to tell apart. Detections match a true box at intersection over union ≥ 0.5. Settings were not varied: $\alpha = 0.05$, stride 4, 27 window shapes, $n_{\text{eff}} = 20$, Wald levels 0.01. Timings are per frame, on two CPU cores of a cloud container, include all stages, and are single-run means over the scene.

Protocol. For each scene the threshold is calibrated on 100 object-free frames that follow a 30-frame warm-up; false alarms are counted on the next 100 object-free frames of the same stream; then the 240-frame scene with objects is processed with the same detector. The results below are on 10 scenes (seeds 2000–2009) that were generated after the last change to the algorithm. The development history, including an earlier run on other scenes that exposed a defect, is in Appendix A.

5.2 Exactness of the window likelihood

On random data ($d = 3$, window 37×31), the log-likelihood from integral images and the direct sum differ by 9×10^{-13} on a value of about -6156, which is floating-point precision; this is also an automated test. Building the nine integral images (three channels, six products) of a 640×480 frame takes about 8 ms in our environment.

5.3 False alarms

Table 2 (column "False-alarm rate") gives the share of the 100 object-free test frames that contain at least one detection, averaged over the 10 scenes. With the calibrated threshold no frame raised an alarm in 1000 frames, against a target of $\alpha = 0.05$ (one-sided 95% upper bound 0.003 for the stream, Section 4). This does *not* show that the bound of Proposition 2 holds: the calibration frames lie early in the stream, when the background has just been initialised from 30 frames, so their maxima are larger than later ones and the threshold is too high. Section 5.6 measures this directly and finds an exceedance of 0.000 in this protocol and of 0.100 if the threshold is instead calibrated after 100 burn-in frames. A conventional fixed threshold of $Z > 3$ is useless here: it raises an alarm in every frame (rate 1.000) and its precision falls to 0.06. Without the illumination compensation of Section 3.2 the rate is 0.53: the slow drift of the scene is read as motion by large windows.

Method	Precision	Recall	F_1	False-alarm rate	ms/frame
SWIFT-MO (calibrated)	0.873 ± 0.085	0.779 ± 0.112	0.823 ± 0.099	0.000	5.3 ± 0.5
fixed threshold Z>3	0.063 ± 0.009	0.777 ± 0.111	0.116 ± 0.016	1.000	20.3 ± 2.6
no illumination compensation	0.859 ± 0.099	0.770 ± 0.120	0.811 ± 0.109	0.527	4.4 ± 0.5
no overlap suppression	0.026 ± 0.007	0.974 ± 0.017	0.050 ± 0.013	0.000	115.4 ± 66.5
MOG2	0.926 ± 0.051	0.793 ± 0.107	0.853 ± 0.085	0.000	1.7 ± 0.3
KNN	0.926 ± 0.051	0.792 ± 0.107	0.853 ± 0.085	0.000	2.3 ± 0.3

Table 2: Detection on 10 held-out synthetic scenes (mean ± sd over scenes). SWIFT-MO rows are the full method and three ablations; MOG2 and KNN are OpenCV background subtractors followed by a morphological opening and connected components (minimum area 150 pixels). The baselines give no class labels.

Metric	Mean difference	95% t-interval	Wilcoxon p
Precision	-0.053	[-0.098, -0.007]	0.020
Recall	-0.014	[-0.053, 0.026]	0.557
F_1	-0.030	[-0.073, 0.012]	0.193
ms per frame	3.56	[3.22, 3.90]	0.002

Table 3: SWIFT-MO (calibrated) minus MOG2, paired over the 10 held-out scenes.

5.4 Detection compared with background subtraction

Table 2 compares the methods. The full method reaches F_1 0.82 (±0.10 over scenes); MOG2 and KNN reach 0.85 and run 2 to 3 times faster (2.3 and 1.7 ms against 5.3 ms per frame). So on these scenes, for detection alone, the scan statistic is not better than pixel-wise background subtraction and costs more. Removing overlap suppression is catastrophic (precision 0.03, about 100 ms per frame), because each object is then reported by dozens of overlapping windows.

Table 3 gives the paired differences. Against MOG2, the loss in F_1 (-0.030) is not significant with ten scenes (95% interval from -0.073 to $+0.012$, Wilcoxon $p = 0.19$), so the data are compatible with equal detection quality; the loss in precision (-0.053) is significant at the 5% level (without correction for the four metrics examined), recall does not differ, and the extra time per frame of 3.6 ms is clearly different from zero. The scenes are synthetic and share one generator, so the intervals describe variation between scenes of this generator only.

5.5 Sequential classification

Table 4 compares the sequential test with decisions after a fixed number of frames. The sequential test is never more accurate than the better of the two fixed rules; at separation 1 and 0.25 it decides after about one frame because the evidence accumulated per frame already crosses Wald's boundary, and at separation 0.1 it needs about four frames to reach an accuracy (0.89) close to that of five fixed frames (0.90). Accuracy stays at about 0.82 to 0.90 at all separations, including the easiest one, which indicates that most errors come from earlier stages (tracks that merge or swap objects, mask pixels that belong to the background, tracks fragmented across objects) and not from the class models. We have not isolated these causes, so we claim no benefit of the sequential test here, and its error levels should not be read as guarantees (frames of a track are dependent and the evidence is scaled by n_{eff}).

5.6 Does the guarantee hold? A calibration study

Proposition 2 requires exchangeable maxima. We test this on 30 further object-free streams (seeds 3000–3029) of $30 + B + 100 + 200$ frames: after a 30-frame initialisation, B burn-in frames are discarded, $m = 100$ frames calibrate c ($\alpha = 0.05$, $k = 96$), and 200 later frames measure the

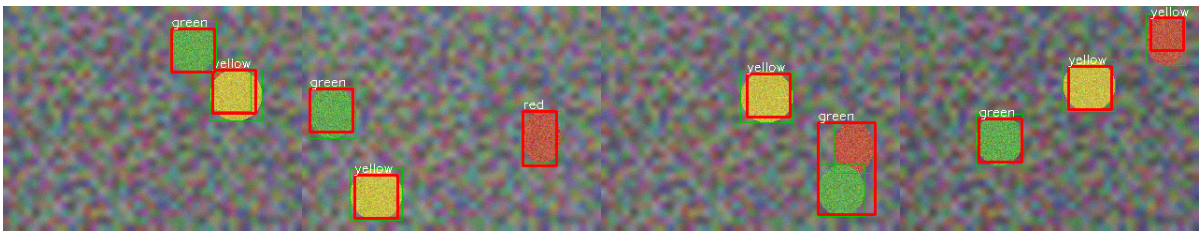


Figure 4: Frames 10, 60, 120 and 180 of the development scene. Green: ground truth; red: detections, labelled with the class once the sequential test has decided. Objects appear as textured ellipses because pixels are independent Gaussian.

Class separation	Tracks	Acc. after 1 frame	Acc. after 5 frames	SPRT decided	SPRT accuracy	SPRT frames
0.1	80	0.887	0.903	0.938	0.893	3.91
0.25	85	0.847	0.895	1.000	0.835	1.11
1	77	0.870	0.870	1.000	0.818	1.01

Table 4: Class decisions per track on the held-out scenes. "Acc. after n frames" is the accuracy of the class with the largest accumulated log-likelihood after n frames of the track; SPRT is Wald's test with levels 0.01. Tracks are matched to the true object they overlap most frames.

achieved exceedance $\widehat{U}$, the share of their maxima above c. We take $B = 0$ (the protocol of the benchmark, mode A), $B = 100$ (mode B) and $B = 300$ (mode C). Under exchangeability $\widehat{U}$ has mean ≤ 0.0495 and, from Proposition 7 and binomial sampling of 200 frames, standard deviation 0.026.

The result (Table 5, Figure 5) is in three parts. (i) Calibrated right after initialisation (A), the threshold is 14.0 against test maxima of at most 7.8 on average, and no test frame ever exceeds it: the method is valid but wastes power, since a threshold near 6.9 would have been enough. (ii) After 100 burn-in frames (B), the exceedance is 0.100 (95% interval 0.085 to 0.115), twice the target: the guarantee is *violated*. The cause is visible in Figure 5: the maxima keep rising slowly after the calibration window as the per-pixel variances adapt, which breaks exchangeability in the anti-conservative direction. (iii) After 300 burn-in frames (C), the exceedance is 0.045 (0.037 to 0.054) with standard deviation 0.023, close to the values 0.0495 and 0.026 expected from Proposition 7 and binomial sampling, so the guarantee holds in practice once the background has settled. The practical rule is therefore to calibrate only after the background statistics have stabilised, and to monitor the exceedance on later object-free frames. These are synthetic streams with a stationary drift; real video with slow scene changes can break exchangeability in either direction.

5.7 A trained detector, for scale only

In an earlier experiment of the same package on a different task (faces in 640×480 video), the UltraLight RFB detector took 13.7 ms per frame and the Haar cascade 100.9 ms on a Colab CPU. The tasks, frame sizes, objects and machines differ, so this is not a comparison. No trained detector has been run on the scenes of this paper.

6 Discussion

What the results support. (i) Window likelihoods from integral images are exact and cheap, so a scan over about 8.5×10^4 windows fits in a few milliseconds per 320×240 frame. (ii) A threshold taken from the null maxima is necessary: a conventional fixed threshold fails completely on the same data. It keeps false alarms at the target only if the calibration frames

Burn-in before calibration	Threshold c	Median M (test)	Max M (test)	Exceedance, mean [95% CI]	sd	max
A: 0 frames	14.00	5.50	7.79	0.000 [0.000, 0.000]	0.000	0.000
B: 100 frames	6.57	5.61	7.85	0.100 [0.085, 0.115]	0.040	0.180
C: 300 frames	6.89	5.62	7.71	0.045 [0.037, 0.054]	0.023	0.095
Prediction under exchangeability	–	–	–	0.050	0.026	–

Table 5: Calibration study, 30 streams per row. Threshold and median/maximum of the scan maxima M on the test frames are means over streams; the exceedance is the share of the 200 test maxima above the threshold.

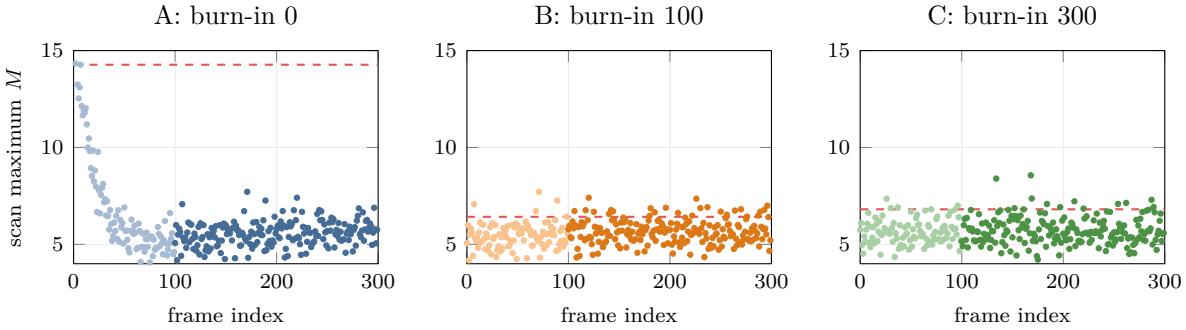


Figure 5: Scan maxima of stream 3000 under the three calibration modes. Light points: the $m = 100$ calibration frames; dark points: the 200 test frames; dashed: the threshold c. In mode A the calibration maxima are larger than the test maxima (the background is still settling), so c is far above anything seen later; in mode B the maxima drift upward, so the threshold is too low for the test frames.

are exchangeable with later ones: it did after 300 burn-in frames (0.045 against 0.05), it was twice too permissive after 100 (0.100), and it was very conservative when calibrated at once after initialisation (0.000). (iii) Compensation for global illumination matters when the scene drifts. (iv) Overlap suppression is essential.

What they do not support. The method is not more accurate than MOG2/KNN for detection and is slower; the sequential test brought no gain in accuracy; the guarantee was checked only on stationary synthetic streams, and it failed when the calibration window was too early. The only advantages left are the class labels that the baselines do not provide and the calibrated, interpretable threshold, and the former was shown only on colour-coded synthetic objects.

Assumptions that can fail on real video.

- *Pixel model.* Independent Gaussian pixels are wrong for natural images (correlated noise, texture, shadows); then Z is not centred and scaled as in (2). Proposition 2 does not need the model to be right, which is why the threshold is calibrated, but a wrong model lowers power.
- *Exchangeability.* Slow drift, camera shake and periodic motion (waves, foliage) violate it, and an adaptive background makes consecutive frames dependent.
- *Background absorption.* Objects that stop are learned as background; shadows count as motion.
- *Class models.* Colour alone cannot separate classes of the same colour. Gradients or texture can be added as channels, at the cost of more integral images; not tested.

- *Tracking.* Overlap-based association fragments tracks; a motion model (Kalman filter) would be a natural improvement.
- *Window set.* A coarse grid and three aspect ratios limit localisation; very small or elongated objects are missed.

Work needed to turn this into a contribution. (1) Real video: CDnet 2014 categories, with the object-free segments used to check the false-alarm bound. (2) A trained lightweight detector with a tracker, on the same sequences and one CPU thread. (3) Temporal gating: scan all windows every m frames and only a neighbourhood of predicted boxes otherwise; it is the natural way to gain speed, but it is not implemented, so no gain is claimed. (4) A tracker with a motion model, then re-measurement of the class accuracy to see whether the sequential test helps once upstream errors are removed. (5) Correlated-noise and camera-jitter experiments for the calibration, and an online test of exchangeability (for instance a drift test on the stream of maxima) that decides when calibration may start. (6) A systematic literature search for prior work.

7 Conclusion

SWIFT-MO scores every sub-window of a frame in constant time from integral images of sufficient statistics, sets its alarm threshold by a finite-sample rank argument that covers all windows at once, and classifies tracked objects sequentially. On ten held-out synthetic scenes it scans 84,550 windows per 320×240 frame in 5.3 ms and detects objects with F_1 0.82, slightly below and slower than OpenCV's background subtractors, which give no classes. A separate calibration study shows that the false-alarm bound holds in practice (0.045 against 0.05) when the threshold is calibrated after the background has settled, fails by a factor two after a short burn-in, and is conservative when calibrated at once. The calibrated threshold and the illumination compensation are shown to be necessary on these data; the sequential classifier is not shown to help. Whether the properties survive real video, and how the method compares with trained detectors, remains open, and the experiments that would decide it are listed in Section 6.

Reproducibility

The package `fastfacetrack` (module `swift`, `moving`, `benchmark`) is at `https://github.com/yamenetoo/Fast-Face-tracking`. The tables of this paper are produced by `paper/make_tables.py` and `paper/make_stats.py` from `results/benchmark/*.csv` and `results/calibration/*.csv` (the latter by `fastfacetrack.benchmark.run_calibration_study`, about 40 minutes); the benchmark is `fastfacetrack.benchmark.run_benchmark` (about 12 minutes on two CPU cores).

A Development history and the earlier run

The algorithm was developed on one synthetic scene (seed 0) and on scenes with seeds 50–54 used for false-alarm tests. Three changes were made after looking at results on that scene: (a) local-maximum selection per window size and overlap suppression, after an early version had precision 0.14 and recall 0.68; (b) a false-alarm test that calibrates on the same stream, after a first design that calibrated on one scene and tested on others (which Proposition 2 does not cover) gave 0.10; (c) the illumination compensation.

A first held-out run on ten scenes (seeds 1000–1009) then gave the results of Table 6. It exposed a defect: the variance update used the uncompensated difference, so a slow global drift

inflated the background variance, which lowered both the null maxima and the power. In that run the fixed threshold $Z > 3$ looked as good as the calibrated one (false-alarm rate 0), which was an artefact of the inflated variance, not a property of the method. We corrected the update (the variance now uses the compensated difference) and ran a second held-out evaluation on ten new scenes (seeds 2000–2009), which is the one reported in Section 5. Both runs are kept in the repository (`results/benchmark_v1`, `results/benchmark`).

Method	Precision	Recall	F_1	False-alarm rate	ms/frame
SWIFT-MO (calibrated)	0.892 ± 0.086	0.815 ± 0.137	0.850 ± 0.111	0.000	4.7 ± 0.5
fixed threshold Z>3	0.892 ± 0.086	0.815 ± 0.137	0.850 ± 0.111	0.000	5.5 ± 0.5
no illumination compensation	0.882 ± 0.091	0.807 ± 0.136	0.841 ± 0.113	0.529	4.6 ± 0.5
no overlap suppression	0.026 ± 0.005	0.963 ± 0.037	0.050 ± 0.009	0.000	105.1 ± 42.2
MOG2	0.937 ± 0.051	0.813 ± 0.136	0.868 ± 0.101	0.000	1.8 ± 0.2
KNN	0.937 ± 0.051	0.813 ± 0.136	0.868 ± 0.101	0.000	2.4 ± 0.2

Table 6: First held-out run (seeds 1000–1009) with the variance defect. Compare the fixed-threshold row with Table 2.

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